Ziyao Xu

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Dept of Math and Stats
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Education

- **Brown University** — Providence, RI, USA
  Ph.D. in Applied Mathematics. Advisor: Chi-Wang Shu. — *Sep. 2019 – May. 2023*
- **Michigan Technological University** — Houghton, MI, USA
  M.S. in Mathematical Sciences. Advisor: Yang Yang. — *Jan. 2017 – May. 2019*
- **China University of Petroleum** — Qingdao, Shandong, China
  B.E. in Petroleum Engineering. — *Sep. 2012 – Jun. 2016*

Professional Experience

- **Binghamton University** — Binghamton, NY, USA
  Assistant Professor — *Aug. 18, 2025 – Present*
- **University of Notre Dame** — Notre Dame, IN, USA
  Postdoctoral Research Associate — *Jul. 1, 2023 – Aug. 15, 2025*

Research Interests

- Finite Element Methods
- Finite Difference Methods
- Computational Fluid Dynamics
- Flow and Transport in Porous Media

Preprints

1. Ziyao Xu and Yong-Tao Zhang, Exponential Time Differencing Runge-Kutta Discontinuous Galerkin (ETD-RKDG) Methods for Nonlinear Degenerate Parabolic Equations, preprint.
2. Ziyao Xu and Zheng Sun, TVD and TVB Preservation without TVD Time Discretization for Discontinuous Galerkin Methods, preprint.
3. Ziyao Xu, An unfitted finite element discrete fracture model for low-permeability barriers via local stiffness matrix modification, preprint.
4. Ziyao Xu, A non-conforming finite difference discrete fracture model based on an energy principle, preprint.
5. Yuanzhe Wei, Ziyao Xu and Chi-Wang Shu, A well-balanced conservative high-order alternative finite difference WENO (A-WENO) method for Euler equations with gravity, preprint.

Publications

1. Jingyao Liu, Hui Guo, Zhuozheng Chen and Ziyao Xu, A high-order finite volume discrete fracture model for single-phase flow in fractured porous media, *Advances in Water Resources*, v216 (2026), 105425.
2. Yong Liu and Ziyao Xu, An interior penalty discontinuous Galerkin method for an interface model of flow in fractured porous media, *Journal of Scientific Computing*, v107, (2026), 83.
3. Ziyao Xu, Guanyang Liu and Yong-Tao Zhang, A Conservative and Positivity-Preserving Discontinuous Galerkin Method for the Population Balance Equation, *Journal of Scientific Computing*, v106 (2026), 67.
4. Ziyao Xu, Zheng Sun and Yong-Tao Zhang, Stability and Time-Step Constraints of Exponential Time Differencing Runge–Kutta Discontinuous Galerkin Methods for Advection-Diffusion Equations, *Journal of Scientific Computing*, v105 (2025), 59.

5. Xinyu Wu, Hui Guo, Ziyao Xu, Lulu Tian and Yang Yang, A reinterpreted discrete fracture model for wormhole propagation in fractured porous media, *Journal of Computational Physics*, v532 (2025), 113953.

6. Ziyao Xu and Yong-Tao Zhang, High-order exponential time differencing multi-resolution alternative finite difference WENO methods for nonlinear degenerate parabolic equations, *Journal of Computational Physics*, v529 (2025), 113838.

7. Ziyao Xu and Chi-Wang Shu, A well-balanced conservative high-order alternative finite difference WENO (A-WENO) method for the shallow water equations, *Advances in Water Resources*, v196 (2025), 104898.

8. Ziyao Xu and Dennis Glaser, An extension of the box method discrete fracture model (Box-DFM) to include low-permeable barriers with minimal additional degrees of freedom, *Advances in Water Resources*, v195 (2025), 104869.

9. Ziyao Xu and Chi-Wang Shu, A high-order well-balanced discontinuous Galerkin method for hyperbolic balance laws based on the Gauss-Lobatto quadrature rules, *Journal of Scientific Computing*, v101 (2024), 39.

10. Ziyao Xu and Chi-Wang Shu, Local characteristic decomposition-free high order finite difference WENO schemes for hyperbolic systems endowed with a coordinate system of Riemann invariants, *SIAM Journal on Scientific Computing*, v46 (2024), pp.A1352-A1372.

11. Xinyu Wu, Hui Guo, Ziyao Xu and Yang Yang, A reinterpreted discrete fracture model for Darcy-Forchheimer flow in fractured porous media, *Advances in Water Resources*, v179 (2023), 104504.

12. Ziyao Xu and Chi-Wang Shu, On the conservation property of positivity-preserving discontinuous Galerkin methods for stationary hyperbolic equations, *Journal of Computational Physics*, v490 (2023), 112304.

13. Ziyao Xu, Zhaoqin Huang and Yang Yang, The hybrid-dimensional Darcy's law: A non-conforming reinterpreted discrete fracture model (RDFM) for single-phase flow in fractured media, *Journal of Computational Physics*, v473 (2022), 111749.

14. Ziyao Xu and Chi-Wang Shu, Third order maximum-principle-satisfying and positivity-preserving Lax-Wendroff discontinuous Galerkin methods for hyperbolic conservation laws, *Journal of Computational Physics*, v470 (2022), 111591.

15. Ziyao Xu and Chi-Wang Shu, High order conservative positivity-preserving discontinuous Galerkin method for stationary hyperbolic equations, *Journal of Computational Physics*, v466 (2022), 111410.

16. Hui Guo, Wenjing Feng, Ziyao Xu and Yang Yang, Conservative numerical methods for the reinterpreted discrete fracture model on non-conforming meshes and their applications in contaminant transportation in fractured porous media, *Advances in Water Resources*, v153 (2021), 103951.

17. Ziyao Xu and Yang Yang, The hybrid dimensional representation of permeability tensor: A reinterpretation of the discrete fracture model and its extension on nonconforming meshes, *Journal of Computational Physics*, v415 (2020), 109523.

18. Ziyao Xu, Yang Yang and Hui Guo, High-order bound-preserving discontinuous Galerkin methods for wormhole propagation on triangular meshes, *Journal of Computational Physics*, v390 (2019), pp.323-341.

19. Nattaporn Chuenjarern, Ziyao Xu and Yang Yang, High-order bound-preserving discontinuous Galerkin methods for compressible miscible displacements in porous media on triangular meshes, *Journal of Computational Physics*, v378 (2019), pp.110-128.

20. Hui Guo, Lulu Tian, Ziyao Xu, Yang Yang and Ning Qi, High-order local discontinuous Galerkin method for simulating wormhole propagation, *Journal of Computational and Applied Mathematics*, v350 (2019), pp.247-261.

Conference Proceedings

1. Ziyao Xu and Yang Yang, The hybrid-dimensional Darcy's law: A non-conforming reinterpreted discrete fracture model (RDFM) for the compressible miscible displacement and multicomponent gas flow in fractured media, *SPE Reservoir Simulation Conference*, (2023), SPE-212164-MS.

Presentations (Selected)

- Midwest Numerical Analysis Day 2026, University of Notre Dame, April 1112, 2026.
  Presentation: A Conservative and Positivity-Preserving Discontinuous Galerkin Method for the Population Balance Equation.
- 2025 SIAM New York-New Jersey-Pennsylvania Section Conference, Pennsylvania State University, State College, PA, 31 October - 2 November, 2025.
  Presentation: Session 1: A Conservative and Positivity-Preserving Discontinuous Galerkin Method for the Population Balance Equation; Session 2: Stability and Time-Step Constraints of Exponential Time Differencing Runge-Kutta Discontinuous Galerkin Methods for Advection-Diffusion Equations.
- The 10th Annual Meeting of SIAM Central States Section, University of Arkansas, Fayetteville, Arkansas, October 1112, 2025.
  Presentation: A Conservative and Positivity-Preserving Discontinuous Galerkin Method for the Population Balance Equation.
- Texas-Louisiana Section of SIAM 8th Annual Meeting, University of Texas at Austin, Austin, Texas, September 26-28, 2025.
  Presentation: High-order exponential time differencing (ETD) multi-resolution alternative finite difference WENO methods (A-WENO) for nonlinear degenerate parabolic equations.
- SIAM Southeastern Atlantic Section (SEAS), The University of Tennessee, Knoxville, Mar 21-23, 2025.
  Presentation: Stability and Time-Step Constraints of Exponential Time Differencing Runge-Kutta Discontinuous Galerkin Methods for Advection-Diffusion Equations.
- The third North American High Order Methods Conference (NAHOMCon), Dartmouth College, Hanover, New Hampshire, June 17-19, 2024.
  Presentation: An Interior Penalty Discontinuous Galerkin (IPDG) Method for an Interface Model of Flow in Fractured Porous Media.
- The 8th Annual Meeting of SIAM Central States Section, Lincoln, Nebraska, October 7-8, 2023.
  Presentation: A high-order well-balanced discontinuous Galerkin method for hyperbolic balance laws with non-hydrostatic equilibria.
- SPE Reservoir Simulation Conference, Galveston, Texas, March 28-30, 2023.
  Poster: The Hybrid-Dimensional Darcy's Law: A Non-Conforming Reinterpreted Discrete Fracture Model (RDFM) for the Compressible Miscible Displacement and Multicomponent Gas Flow in Fractured Media.
- 2022 SIAM Annual Meeting, July 11-15, 2022. Co-organizer of MS8, MS23, MS49: Recent Advances in Computational Geosciences.
  Presentation: A Reinterpreted Discrete Fracture Model for Fracture and Barrier Networks on Non-Conforming Meshes.
- Copper Country Workshop on Applied Mathematics, Statistics, and Data Sciences, Michigan Technologycal University, Houghton, MI, July 5-7, 2022.
  Presentation: High Order Conservative Positivity-Preserving Discontinuous Galerkin Method for Stationary Hyperbolic Equations.
- 2021 SIAM Annual Meeting, July 19-24, 2021. Organizer of MS9: Models and Numerical Methods in Computational Geosciences.
  Presentation: The Hybrid-Dimensional Darcy's Law: A Novel Discrete Fracture Model for Fracture and Barrier Networks on Non-Conforming Meshes.

Teaching Experience (Recent)

- MATH 330 Number Systems, Instructor, Binghamton University, Spring 2026
- MATH 478 Real Analysis I, Instructor, Binghamton University, Fall 2025
- ACMS 30530 Introduction to Probability, Instructor, University of Notre Dame, Spring 2025
- ACMS 40630/60630 Nonlinear Dynamical Systems, Instructor, University of Notre Dame, Fall 2023, Fall 2024
- ACMS 20620 Applied Linear Algebra, Instructor, University of Notre Dame, Spring 2024
- APMA 0350 Applied Ordinary Differential Equations, TA, Brown University, Spring 2021
- APMA 1170 Introduction to Computational Linear Algebra, TA, Brown University, Fall 2020

Honors & awards

- David Gottlieb Memorial Award, Division of Applied Mathematics, Brown University, 2023.
- Outstanding Research Award, Department of Mathematical Sciences, Michigan Technological University, 2019
- The First Prize Scholarship, China University of Petroleum, 2015
- National Scholarship, China, 2013, 2014

Skills

- Programming: Matlab, Julia, Python, C, LaTeX
- Languages: English (proficient), Mandarin (native)